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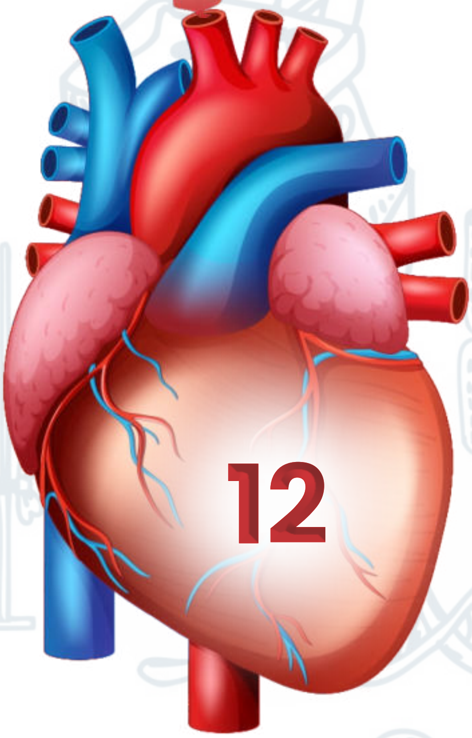

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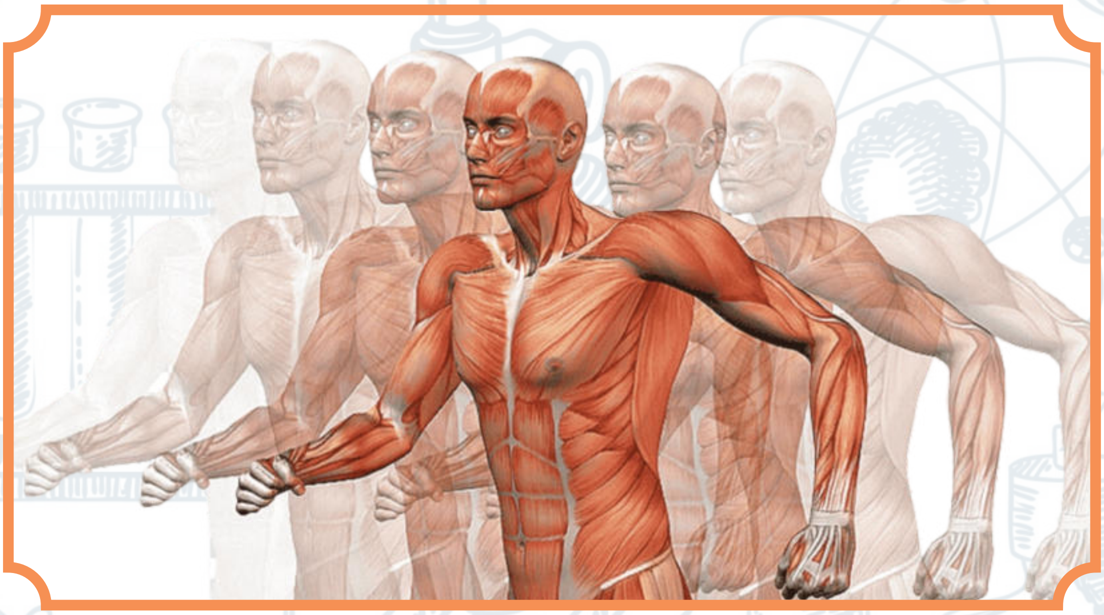
CIRCULATION



CIRCULATION:

The continuous movement of blood through the heart and blood vessels whichis maintained cheifly by the action of heart, and by which nutrients oxygen,and internal secretions carried to and wastes are carried from the body tissue.

TYPES OF CIRCULATORY SYSTEM:

There are two types of circulatory system named below

a) Open type circulatory system.

b) Close type circulatory system.

OPEN CIRCULATORY SYSTEM:

In an open type circulation system the blood mixes with the internal organ directly. The organisms that have that type of circulation do not have a true heart and capillaries. Instead of that they have blood vessels that can act as pumps to force the blood along. Instead of capillaries the blood vessels direct connected with the open sinuses and blood came in direct contact with the tissues. In such organisms there is no blood instead of blood they have a fluid known as **hemolymph.** Ostia is the opening which act as a valve, it can maintain the flow of hemolymph.

Example: Grasshopper, Cockroach.

CLOSE CIRCULATORY SYSTEM:

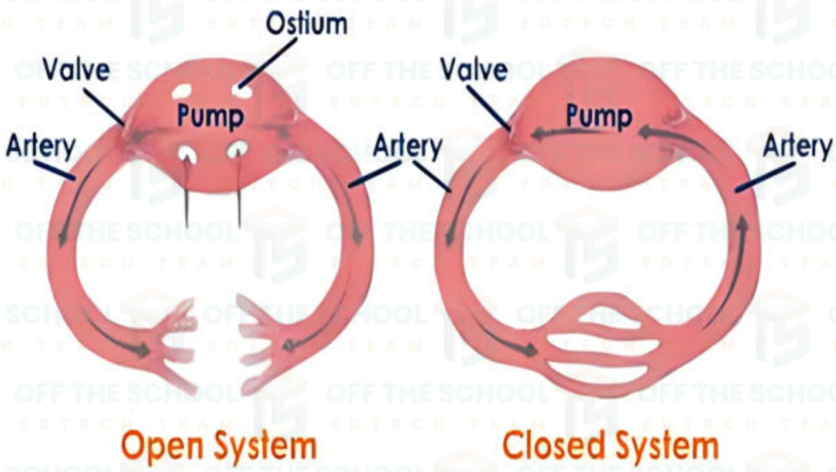
The close circulatory system, blood is pumped through a close system of arteries, veins, and capillaries. Blood does not come in to direct contact with the tissues. Arteries carry blood from the heart. It is then circulated through the microscopic capillaries. Where exchange of material take place. Then it is collected by the veins and transferred to the heart again.

Example: Octopus, Earth warm

Then there are further two types of close circulatory system

a) Single circuit plan.

b) Double circuit plan.



SINGLE CIRCUIT PLAN:

The fish heart are of two chambered, consist of an upper atrium and lower ventricle. However, the fish has two accessory chambers. One accessory chambered is thin walled sinus venosus. Which collects the blood and lead in to atrium, the other accessory chamber is the conus arteriosus, an enlargement of the main artery leading out of ventricle. In fish body blood is pumped by the heart to the gills and then from gills it is directly transferred to all parts of the body, and collected again by the heart. In this way this is called single circuit plan.

DOUBLE CIRCUIT PLAN:

An amphibian, bird, reptiles, mammals double circuit plan is present. In these animals heart receives deoxygenated blood from the body and send them into lungs for the oxygenation. When the blood become oxygenated then it is collected by the heart then it is pumped to whole body parts. This is known as double circuit plan

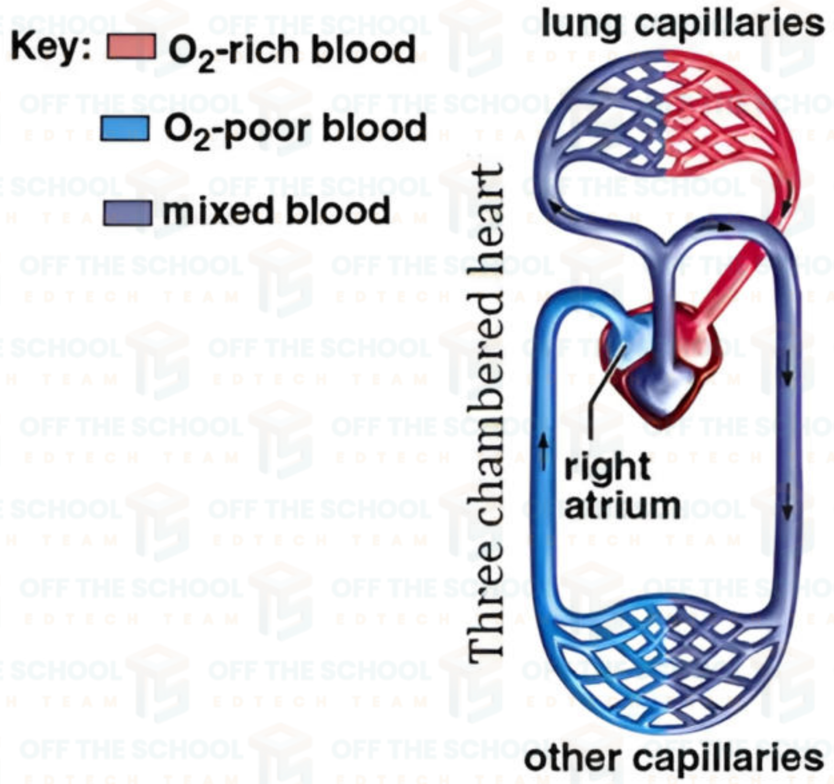
Double circuit plan can further divided in to two types mentioned blow

a) Incomplete double circuit plan.

b) Complete double circuit plan.

INCOMPLETE DOUBLE CIRCUIT PLAN:

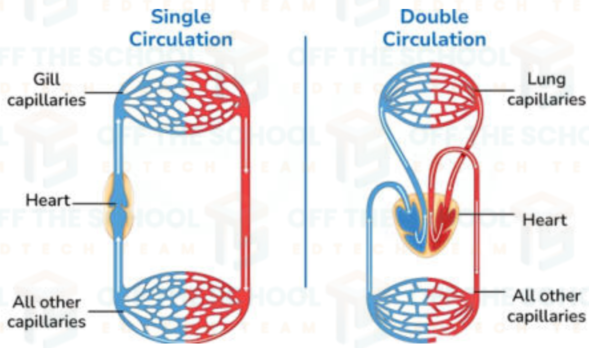
In amphibians and reptiles the heart consist of three chambers. i.e two atrium and one ventricle. The left atrium receives the oxygenated blood and the right atrium receive the



deoxygenated blood. But in the ventricle the deoxygenated and the oxygenated blood mix with each other this is known as incomplete double circuit plan.

COMPLETE DOUBLE CIRCUIT PLAN:

In birds and mammals the heart consist of four chambers. i.e two atrium and two ventricles. The left atrium receive the oxygenated blood and the right atrium receive the deoxygenated blood. So left ventricle receive the oxygenated blood from the left atrium and right ventricle receive the deoxygenated blood from the right atrium. In this way the blood dose not mix with each other this is known as complete double circuit plan.



HUMAN HEART:

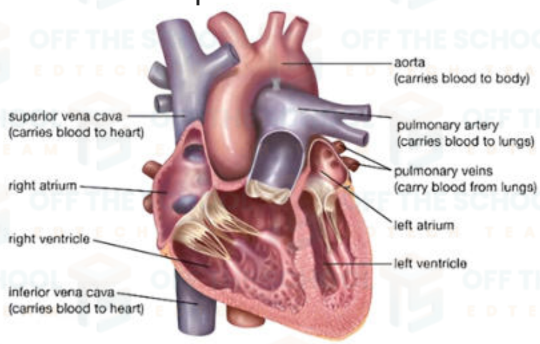
Human heart is most powerful organ in the circulatory system; it is conical hollow and muscular organ. Heart lies in the thoracic cavity between the lungs enclosed in the ribcage. It is ½ on the right sight and 2/3 on the left sight.

MAIN FUNCTION OF HEART:

The heart works continuously like a muscular pump and pump the blood to all the body parts to meet meet their nutritive requirement (O2).

COVERING OF HEART:

Heart is surrounded by the double layer of pericardium. the pericardium is the thick covering enclose heart and the roots of the heart vessels. It consist of three layers. The outer layer is called fibrous layer and also known as fibrous pericardium. It consist of thick connective tissues. An inner double serous membrane called serous pericardium between these two layer there is a small amount of fluid known as serous pericardial fluid which provide mechanical protection.



FUNCTION OF PERICARDIAL FLUID:

pericardial fluid act as lubricant and reduce friction between the heart walls and surrounding tissues during beating of heart.

Layers of heart:

There are three main layers of heart.

a) Epicardium.
b) Myocardium.
c) Endocardium.

Epicardium: it is the outer most layer of heart formed by the visceral layer of pericardium.

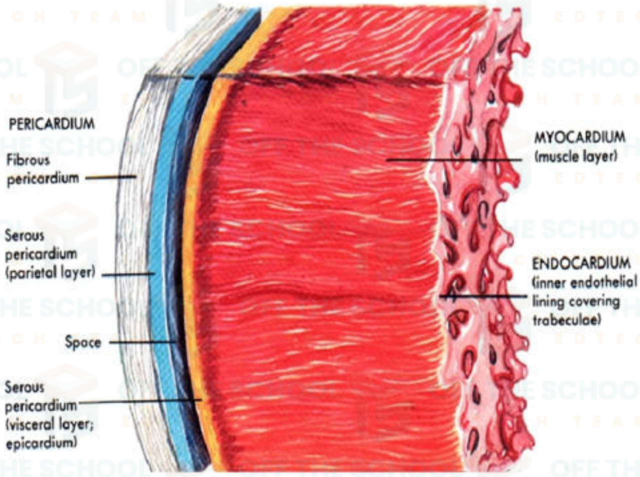
Myocardium: it is the middle muscular layer of heart consist of excited and conductive tissues.

Endocardium: it is the inner most layer of heart and compose of valve, inner lining of chambers, and contains vessels and nerves.

CHAMBERS OF HEART:

There are four chambers of human heart.

a) Right atrium.
b) Right ventricle.
c) Left atrium.
d) Left ventricle.



RIGHT ATRIUM:

Right atrium is the right upper sight chamber of heart and it is thin walled low pressure pump.

Openings of right atrium are

- Superior vena cava.
- Inferior vena cava.
- Coronary sinus.

Function: it receives deoxygenated blood from the whole body and pump it to the right ventricle through the right atrio-ventricular (tricuspid opening) valve.

RIGHT VENTRICLE:

Right ventricle is the right lower chamber of heart, which is triangular in shape

Openings of right ventricle are

- Tricuspid valve.
- Pulmonary artery through pulmonary valve.

Function: right ventricle receives the deoxygenated blood from the right atrium and pumps it to the lungs through pulmonary aorta for oxygenation.

LEFT ATRIUM:

Left ventricle is the upper chamber which is present posteriorly. It receives blood from the lungs.Opening of left atrium are

- Two pairs of pulmonary veins

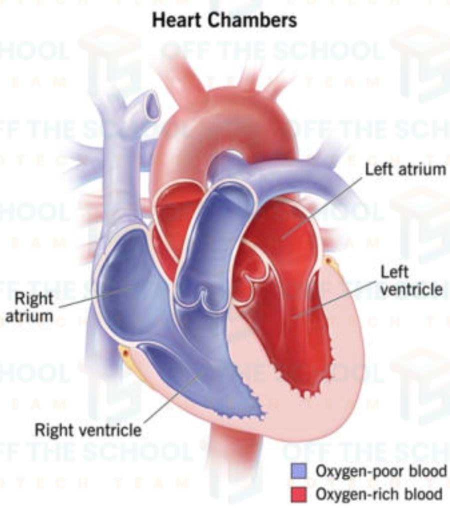
Function: it receives oxygenated blood from the lungs through pulmonary veins

LEFT VENTRICLE:

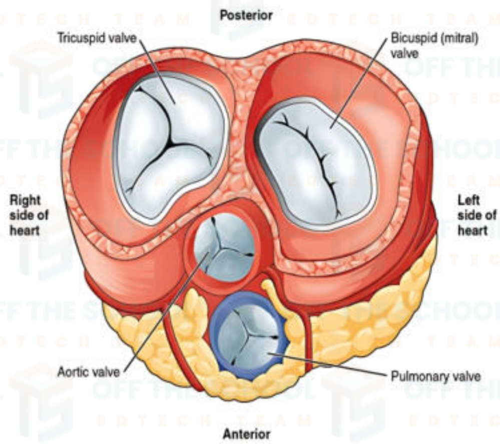
left ventricle is the most thick walled chamber and form the apex of heart.

Opening of left ventricle are

- Bicuspid or mitral valve
- Systemic aorta through aortic valve



Function: it receives the oxygenated blood from the left atrium and pump it in to aorta. So that blood is supplied to all cells of the body.



VALVES OF HEART:
there are two types of valves of heart that are mentioned below.
a) Atrio-ventricular valve.
b) Semilunar valve

ATRIO-VENTRICULAR VALVE:
The valve that present between the artries and the ventricles are called atrio ventricular valve. They are of two type
- Bicuspid valve.
- Tricuspid valve.

BICUSPID VALVE:
Blood flow from left atrium to left ventricle through left atrio-ventricular orifice, which is guarded by the bicuspid or mitral valve. It has two layers of cups so named as bicuspid.

TRICUSPID VALVE:
Blood flow from right atrium to left ventricle through left atrio-ventricular orifice, which is guarded by tricuspid valve. It has three layers of cup so it is called tricuspid.

SEMILUNAR VALVE:
This is the second category of heart valve, which is guarded by the pulmonary and systemic aorta. Types of semilunar valves are
- Aortic valve.
- Pulmonary valve.

PULMONARY VALVE:
This valve guards the pulmonary orifice in left ventricles. It also consist of 3 semilunar valve

AORTIC VALVE:
This valve guards the aortic orifice in left ventricle. It also consist of 3 semilunar cups.

SEPTUM OF HEART:
There are two septum of heart.
- Inter atrial septum.
- Inter ventricular septum.

INTER ATRIAL SEPTUM:

Internally the left and right atrium are separated by this septum. (through vertical membrane)

INTER VENTRICULAR SEPTUM:

The right and left ventricles are also separates by this septum. (through thick membrane)

PHASE OF HEART BEAT:

• The contraction of heart chambers in a systemic and regular manner is known as heartbeat. The average human heartbeat is 72 beats per minute.

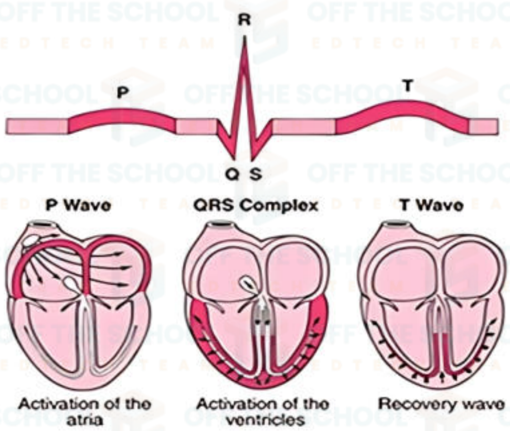
•Heart sounds or heartbeats are the sound generated by the beating heart and the resultant flow of blood through it. 0.8 seconds is time for one heartbeat.

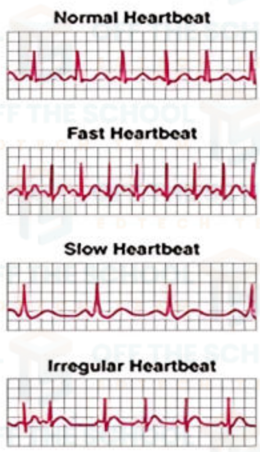
• In healthy adults there are two normal heart sounds often describes as (LUB) and a (DUB) that occur in sequence with each other beat.

•In heartbeat when ventricle contract, blood is forced against the closed Atrio-ventricular valve. It produced first heard sound (LUB) it takes 0.3 seconds after this ventricle show diastole (relaxation). In pulmonary and aorta high pressure is developed diastole relaxation in pulmonary and aorta high pressure is developed. It forces some blood backwards which close the valve of both aorta and pulmonary. It produces sound of (DUB) it takes 0.5 seconds

S.A nodes and A.V nodes:

The heart contain special tissues that produce and send electrical impulse to the heart muscle. It is the impulses that trigger the heart to contract. Each time the heart beats, it sends out an electric-like signal. The heart's electrical signal can be measured with a special machine called an electrocardiogram (ECG).





NORMAL (ECG) WAVE:
a normal ECG makes a specific pattern of three recognizable wave in a cardic cycle. These wave are P-wave, QRS wave and T-wave, P-R interval, S-T segment.

P-WAVE:
It is smaller upward wave that appear first which indicates atrial depolarization systole during which excitation is spreads from SA node to all over atrium. About 0.1 second after P-wave begins Atria contract hence P-wave represent atrial systole.

QRS-WAVE:
It is second wave that begins as a little downward wave but continues as a large upright triangular wave and as downward wave which represent the ventricular depolarization systole.
Just after QRS wave begins ventricles start to contract hence QRS wave represent ventricular systole.

T-WAVE:
It is third smaller wave shaped upward deflection which indicates ventricular repolarization diastole it also represent the beginning of ventricular diastole atrial diastole merge with QRS wave.

P-R INTERVAL:
It represent the time required for an impulse to travel through the atria, AV node and bundle of his to reach ventricles.

ST SEGMENT:
It is measured from the end of S to the beginning of T-wave and represent the time when ventricular fibers are fully depolarized.

APPLICATION OF ELECTROCARDIOGRAM:
- It indicates the rate and rhythm or pattern of contraction of heart.
- It gives a clue about the condition of heart muscles and it used to diagnose heart disorder.

BLOOD VESSELS:
The closed or tube through which transportation medium are blood circulate within body call blood vessels there are three types of blood vessels.
- Arteries.
- Capillaries
- Veins

ARTERIES:
Arteries are blood vessels that carry blood away from heart. Arterial walls are able to expand and contract arteries are further divided into arterioles. Arteries carry oxygenated blood except pulmonary artery. Arteries consists of three layers.

1. Tunica externa.
2. Tunica media.
3. Tunica intima.

TUNICA EXTERNA / ADVENTITIA:
The tunica externa also known as the tunica adventitia, it is the outer most layer of blood vessels, surrounding the tunica media. It is mainly composed of collagen and is supported by external elastic lamina. It is made up of connective tissues.

TUNICA MEDIA:
It is made up of smooth mussels cells and elastic tissues. It lies between the tunica intima on the inside and the tunica externa on the outside. It controls the diameter or cavity lumen of arteries.

TUNICA INTERNA / INTIMA:
The tunica intima is the inner most layer of the arteries. It is made up of layers of endothelial cells. The endothelial cells are in direct contact with the blood flow.

CAPILLARIES:
Capillaries are the microscopic blood vessels. There diameter is about 7-9um. These are very thin walled blood vessels in which change of gases occur. In the capillaries, the wall is only one cell layer thick. Also nutrients, waste, and hormones are exchange across the thin walls of capillaries. Control of blood flow in to capillaries beds it done by nerve controlled sphincters.

VEINS:
Blood leaving the capillary beds flow in to a progressively larger series of venules that in turn join to form veins. Veins carry blood from the heart with the exception of the pulmonary veins, blood in veins is deoxygenated. The pulmonary vein carry oxygenated blood from lungs back to the heart. Venules are smaller veins that gather blood from capillary bed in to veins. The veins have valve that prevent back flow of blood.

ROLE OF ARTERIOL IN VESODILATION AND IN VESOCONTRACTION:
the flow of blood in capillaries beds is determined b the variation in arteriole diameter

• **REDUCTION IN BLOOD TOWARDS SKIN:** in cold season blood vessels of skin are reduced. It is called vasocontraction. Due to this process blood flow towards skin become slow and loss of heat through skin is reduced.

• **INCREASE IN BLOOD FLOW TOWARDS SKIN:** in hot season the blood vessels of skin are dilate and vasodilation occur to released body heat.

VASCULAR PATHWAY:

There are two major pathway for circulation in humans

- Pulmonary circulation.
- Systemic circulation.

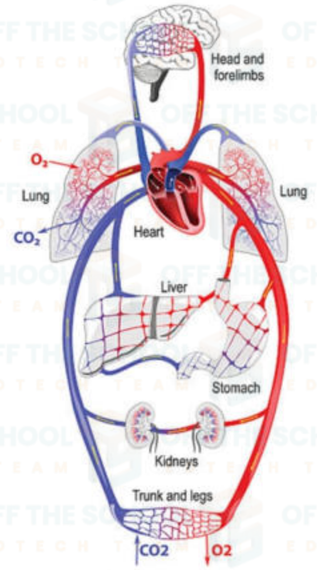
PULMONARY CIRCULATION:

Pulmonary circulation transport oxygen-poor blood from the right ventricle to the lungs, where blood picks up a new blood supply. Then it return the oxygen- rich blood to the atrium.

SYSTEMIC CIRCULATION:

The systemic circulation provides the functional blood supply to all the body tissues. It carries oxygen and nutrient to the cells and pick up carbon dioxide and waste product. Systemic circulation carries oxygenated blood from the left ventricle, through the arteries, to the capillaries in the tissues of the body. From the tissues capillaries, the deoxygenated blood returns through a system of veins to the right atrium of the heart.
It can also further divided in to three major circulation

- Coronary circulation.
- Hepatic portal system.
- Renal circulation.

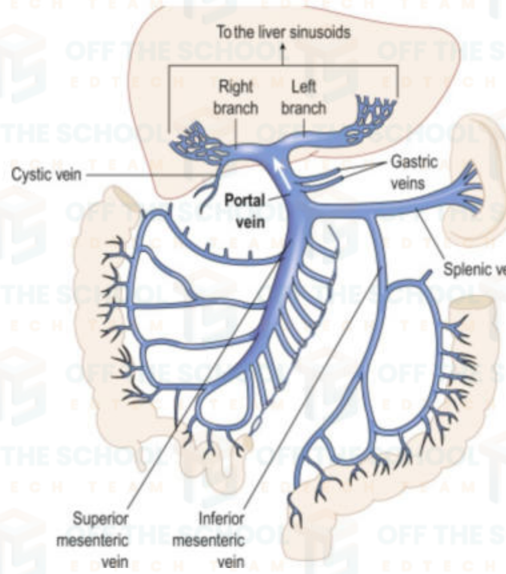


CORONARY CIRCULATION:

It is the circulation of blood in which the blood vessels supply the blood to the heart muscle (myocardium). Coronary arteries supply oxygenated blood to the heart muscle. Cardiac vein than drain away the blood after it has beendeoxygenated.

HEPATIC PORTAL SYSTEM:

The hepatic portal system is a complex system of hepatic portal vein and their capillaries.
The hepatic portal vein is the largest vein the abdominal cavity. It drains blood from the spleen and the gastro-intestinal tract to the liver which contain substance that are absorbed from the small intestine and pass through the liver before going back to the heart. It carries out the main function of further detoxification of deoxygenated blood before reaching the heart.



RENAL CIRCULATION:

Renal circulation supplies the blood to the kidney via the renal arteries, left and right with the branches directly from the abdominal aorta.

Renal artery branches in to arterioles in cortex region that supply blood to the glomeruli.

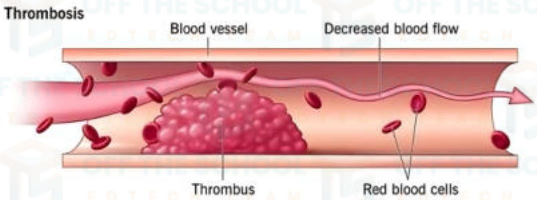
From here blood enter in the peritubular capillaries and vasa recta, then blood is drained through veins and leave the kidney as a single renal vein that empties in to inferior vana cava.

CARDIO-VASCULAR DISORDER:

Disease of heart, blood vessels and blood circulation are generally termed as cardiovascular disorder.

THORUMBUS FORMATION:

This is the formation or presence of a blood clot in a blood vessels. The vessels maybe any artery or vein foe example: in a deep vein thrombosis or a coronary (artery) thrombosis. The clot itself is termed as a thrombus. If the clot breaks loose and travel through the blood stream it is termed as embolus and phenomenon is called embolism.

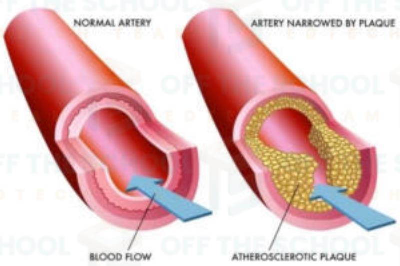


When the blood starts to clot in vessels they lead to the blockage. Initially thrombus block the lumen partially results in decrease blood flow to organ. Later on, thrombus blocks lumen completely so due to complete loss of blood supply, cells damage occur.

ATHEROSCLEROSIS:

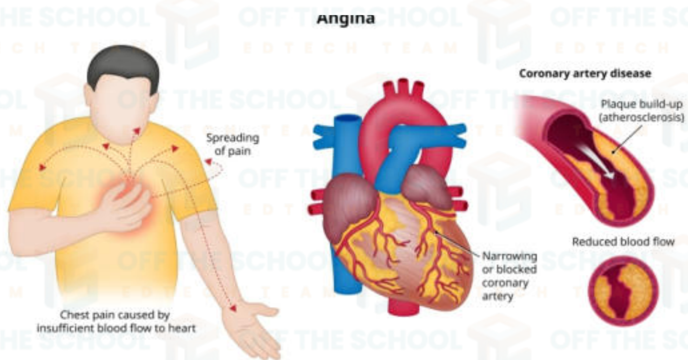
Atherosclerosis is a slow, progressive disease that causes hardening and narrowing of medium to large blood vessels, such as the aorta and the coronary arteries. The narrowing which is due to the formation of fatty lesions called atheromatous plaque in inner lining of arteries. The arteries become hard. Causes are:

- Smoking.
- Obesity.
- Hypertension.
- High cholesterol.
- Diabetes.

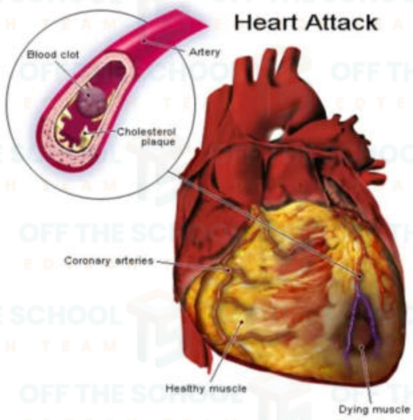


ANGINA PECTORIS:

Angina pectoris is the chest pain or the discomfort that keeps coming back. It happens when some part of your heart dosenot get enough blood and oxygen. angina can be a symptom of coronary artery disease (CAD). This occur when arteries that carry blood to your heart become narrowed and blocked because of atherosclerosis or a blood clot.



HEART ATTACK:



A myocardial infraction occur when the supply of blood and oxygen to an area of the heart muscle is blocked usually by a clot in a coronary artery. Often, thisblockage leads to arrhythmia (irregular heart beat or rhythm) that can cause a severe decrease in the pumping function of the heart and may bring about sudden death. If the blockage is not treated with in a few hours, the affected heart muscle will die.

Sign and symptoms of myocardial infraction:

one of the primary sign of myocardial infraction is chest pain that may also spread to the back shoulders arm, neck, and jaw other symptoms include

- Shortness of breath.
- Nausea.
- Sweating.
- Dizziness.

HEART FAILURE:

Heart failure means that the heart is unable to pump the blood around the body properly it usually happens because the heart has become too weak or stiff.

HYPERTENSION:

hypertension occur when pressure inside the blood vessels is too high blood pressure 160 by 95 MM of HG or above under resting condition is considered to be hypertensive for a man under age 45 a reading above 140 per 95 MM of HG is considered as hypertension.
Hypertension is known as **silent killer** because it does not show any symptom in early stages until a stroke or heart attack occurred
The condition often can be controlled through a combination of lifestyle changes and in many causes, medication
Possible hypertension symptoms
In some cases a person can have symptom hypertension possible symptom associated with hypertension include:

- Dizziness.
- Blurred vision.
- Nausea.
- Headache.

HYPOTENSION:
Hypotension is referred as low blood pressure it has the reading of less than 90 per 60 MM of HG it does not always causes symptoms but you may need treatment if it does. Symptom include dizziness feeling sick blurred vision generally feeling weak confusion and fainting.

CONGENTIAL HEART DISEASE:
Congenital heart disease is a general term for a range of birth defect that affect the normal way the heart works the term congenital means the condition that is permanent from birth.

TREATMENT OF CARDIOVASCULAR DISORDERS:

CORONARY BYPASS:

Coronary bypass surgery redirects blood around a section of a block or partially blocked artery in your heart the procedure involves taking a healthy blood vessel from your left arm or chest and connecting it below and above block arteries in your heart when a new pathway blood flow to the heart muscle improves.

ANGIOPLASTY:

Angioplasty is the procedure use to open block coronary arteries caused by coronary artery disease it restores blood flow to the heart muscle without open heart surgery for angioplasty a long thin tube is put into blood vessel and guided to the blocked coronary artery. The catheters has a tiny balloon in its tip once the catheters is in place the balloon is inflated at the narrowed area of the heart artery this process is placing or blood clot against the side of artery making more room for blood flow.

OPEN HEART SURGERY:

Heart surgery is the surgery that done in the muscles of heart, valves, arteries, or the aorta and the other large arteries connected to the heart the term open heart surgery means that you are connected to the heart lung bypass machine or bypass pump during surgery your heart is stopped while you are connecte to the machine this machine does not work for your and long while your heart is supported for the surgery this machine add oxygen to your blood moves blood through your body and remove carbon dioxide.